**General comments on EE2-6 Control Engineering paper 2016**

Overall, the students have done very well on the exam paper, scoring an exam average of 66%.

1. The students have done relatively well on this question, scoring approximately 68%.
   (a) This is an op–amp modeling and proportional controller design question and is a variation on typical study group questions.
      i. Typical study group question, tests understanding of transfer functions for simple circuits.
      ii. Typical study group question.
      iii. Typical study group question.
      iv. Design part, relatively simple since the design involves a proportional controller.

   (b) This is a Nyquist diagram analysis and design question and is somewhat typical of study group questions, though the system is unstable.
      i. Typical study group question.
      ii. Typical study group question.
      iii. Typical study group question, a bit tricky since it requires an understanding that unstable systems require a minimum value of gain for closed–loop stability.
      iv. Qualitative analysis and design question. The students need to demonstrate an understanding of the compromise between transient and steady–state performance.

2. This is a Nyquist diagram analysis and design question that involves a system with a non–minimum phase factor. The students did well on this question scoring approximately 66%.
   (a) A typical study group question, requires the students to also demonstrate an understanding of the Routh–Hurwitz stability criterion.
   (b) Standard study group question, requires the students to relate the stability margins to both robustness and transient performance.
   (c) Standard Nyquist analysis study group question. Requires an understanding of the relation between the Nyquist diagram and steady–state performance.
   (d) The question requires an analysis of the Nyquist diagram to decide which aspects of the performance require improvement (stead–state versus transient) and to decide what class of controller can achieve that.

3. This is a root-locus type design question for a second order system. The difficulty the students found involves the first specification, which requires designing a PD controller such that the breakaway point of the compensated system is at a given location, which they have to deduce from the second specification. The students did less well on this question, scoring approximately 63%.
   (a) Standard study group question.
   (b) This part explores the limits of what can be achieved by a proportional compensator.
      i. Standard study group question.
      ii. Standard study group question.
   (c) The first part is standard, but for the second part the student need to realise that a critically damped response requires the location of the poles to be at the breakaway points.
   (d) This is the main dynamic controller design part whose difficulty was explained earlier.
   (e) Standard study group question.
   (f) Standard study group question.